Seminar11

luni, 17 ianuarie 2022 10:04

$L \notin TA$; $L \in 2^{\Sigma^*} - TA$

$L_{finite} \subset LR \subset LIC \subset LDC \subset TD \subset TA$

LTD.: Un limbaj L este TD dc. $\exists$ o MT M a.î. pt. $\forall$ $w \in \Sigma^*$, M se termină și scrie Y pe bandă dc. $w \in L$ și M se termină și scrie N pe bandă dc. $w \notin L$.

LTA: Un limbaj L este TA dc. $\exists$ o MT M a.î. pt. $\forall$ $w \in \Sigma^*$, M se termină dc $w \in L$ și M nu se termină dc. $w \notin L$.

Mașina Turing Universală

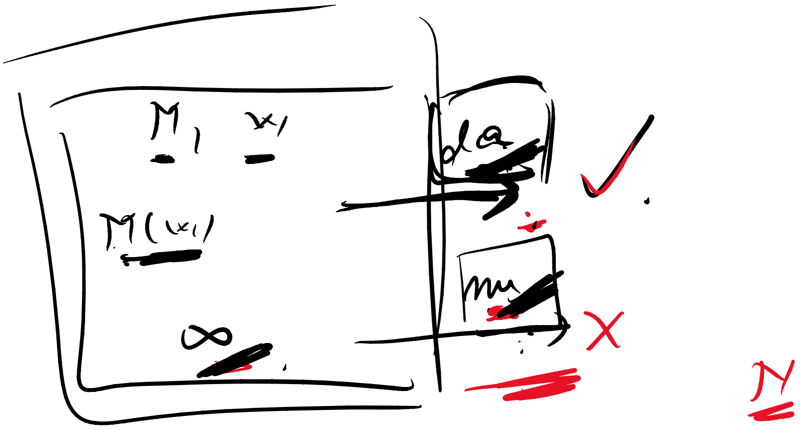
- primește
  - codificarea unei MT $|\rho(M)|$
  - codificarea unui șir $|\rho(w)|$
- execută mașina M ca având w la intrare

1. $L = \{ \rho(M)\, \rho(w) /$ M se oprește cu w la intrare$\}$ $\in TA$.

$\in TD$.

$\in TA$.

$\notin TA$.

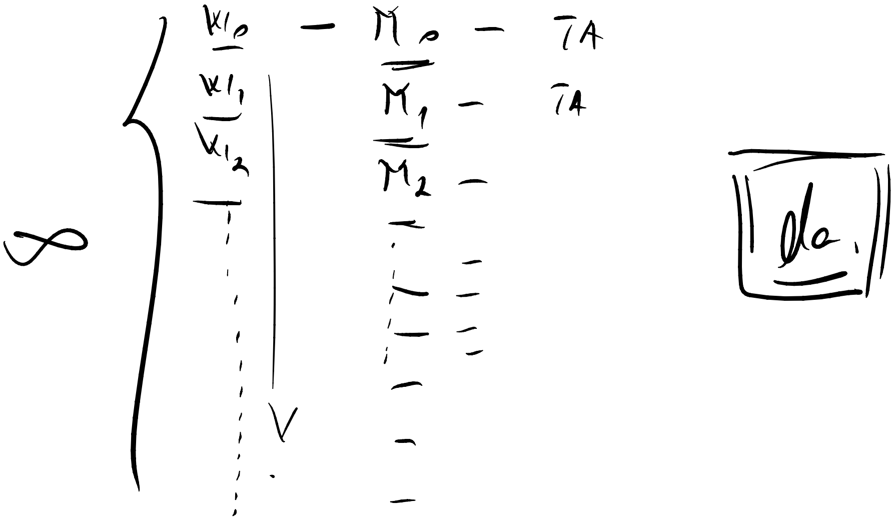


2. $L = \{ \rho(M) /$ pt. $\forall\, w \in \Sigma^*$,

M se oprește cu w la intrare$\}$

$\notin TA$.

Σ^* infinită



Fie: 1. $L \in TD$.

L_1: $M_1 \ \vdash\!\!\xrightarrow{*} Y$, $w \in L_1$

Fie $L_1, L_2 \in TD$.

$L_1 \cup L_2 \in TD$.

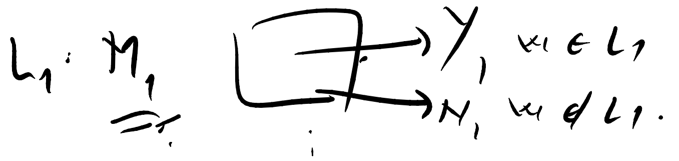
$L_1 \cap L_2 \in TD$.

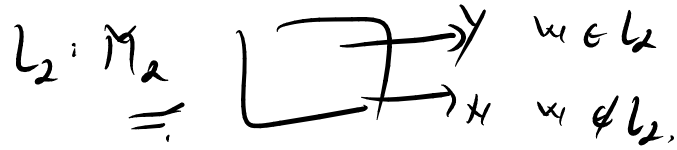
$L_1 \circ L_2 \in TD$

$L_1^* \in TD$

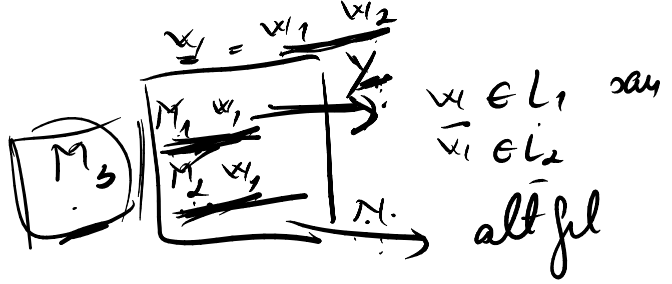
$L_1 \setminus L_2 \in TD$.

$\complement L_1 \in TD$.





$L_3 = L_1 \cup L_2$



Fie $L_1, L_2 \in TA$

$L_1 \cup L_2 \in TA$

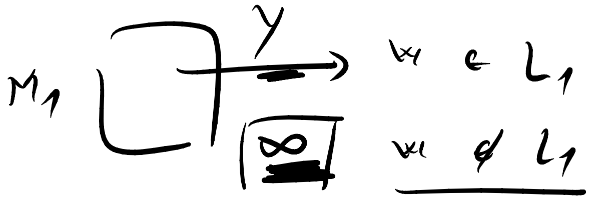
$L_1 \cap L_2 \in TA$

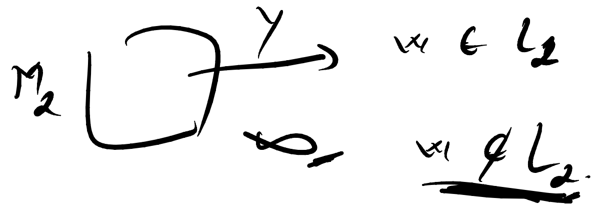
$L_1 \circ L_2 \in TA$

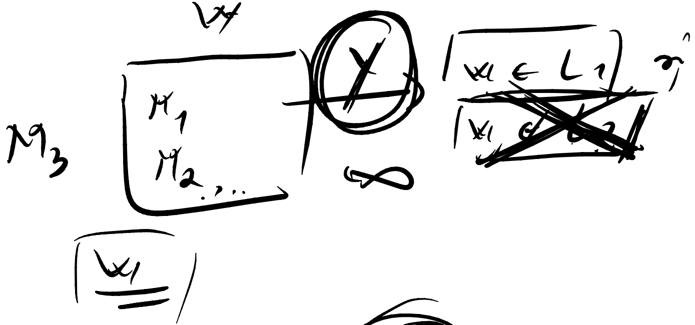
$L_1^* \in TA$

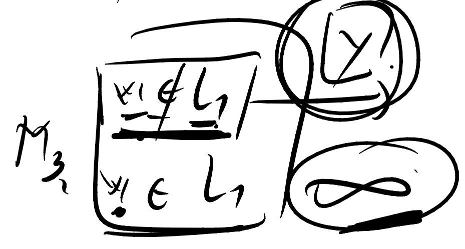
$L_1 \setminus L_2$

$\complement L_1$



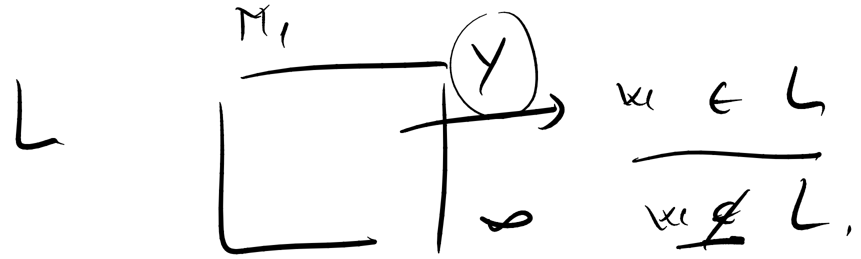


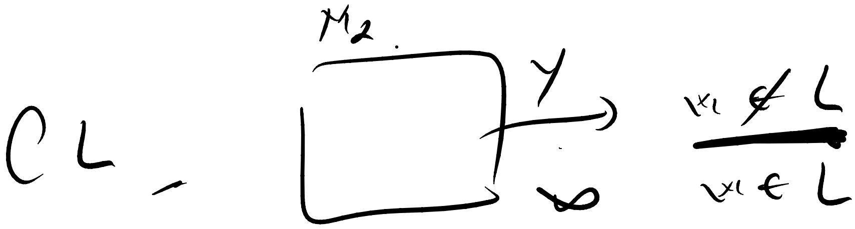


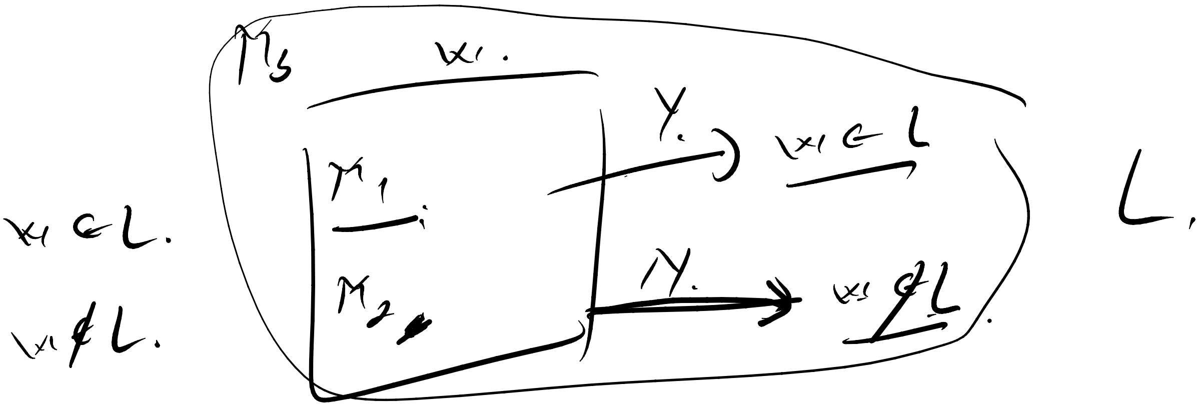


$$\boxed{\text{Dc. } L \in TA \setminus TD \Rightarrow CL \notin TA}$$

Dc. $\dfrac{L \in TA \quad \wedge \quad CL \in TA}{} \Rightarrow L \in TD.$







$\forall L \in LR \Rightarrow L \in LIC.$

$\boxed{L \in LIC} \setminus LR$.

$L \in \underline{TA} \setminus TD$

$TD \subset TA$.

$$\boxed{\text{Dc. } L_1, L_2 \in TA \setminus TD \Rightarrow L_1 \setminus L_2 \notin TA}$$

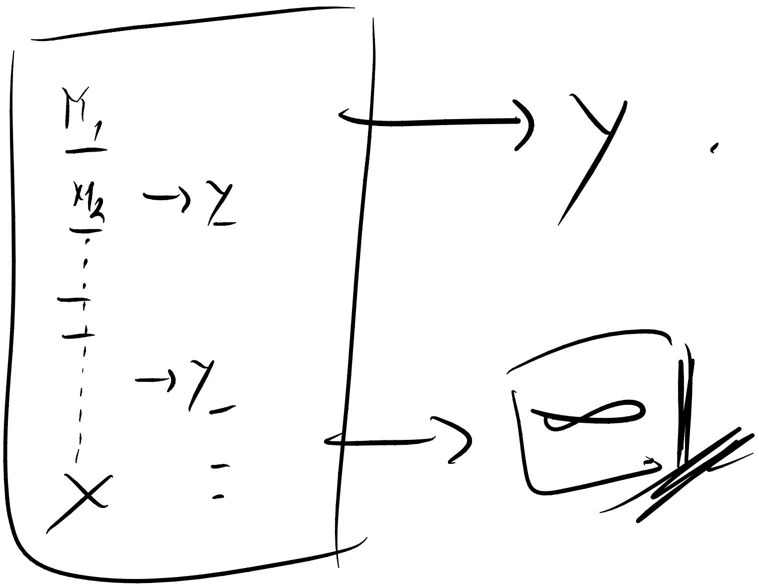
3. $L = \emptyset$

$L \in TD$.

4. $L = \{ \rho(\underline{M}) \,/\, \underline{|L(M)|} \geq 10 \}$. $\in \underline{TA}$.

Ne generăm pas cu pas cuvinte din Σ^*.

Pt. fiecare cuvânt w rulez în paralel M pe w. ----



TA.

Σ^*

$e \to M_1$

$0 \to M_2$

$1 \to$

5. $L = \{ \rho(M) \,/\, \underline{|L(M)| \leq 10} \} \notin TA.$

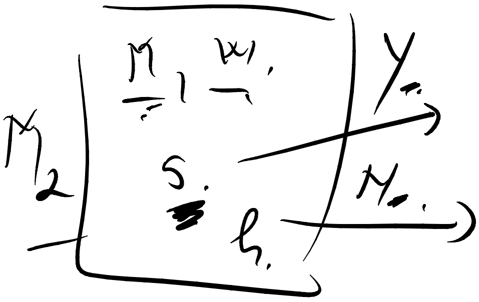
$C\,TA \to \notin TA.$

$\boxed{!!}$

6. $L = \{ \rho(\underline{M}) \rho(\underline{w}) \,/$ M face cel mult $\underline{5}$ tranziții cu

mult 5 / tranziții / cu

w intrare}. ∈ TD



7. $L = \{ \rho(M) /$ ∃ w a.î.
M face cel mult
5 tranziții }.

8. $L = \{ \rho(M)\, \rho(w) /$ w este cel
mai scurt șir
acceptat de M }. ∉ TA

11.
$|w| = 2.$

$\Sigma = \{1\}$

0
1.

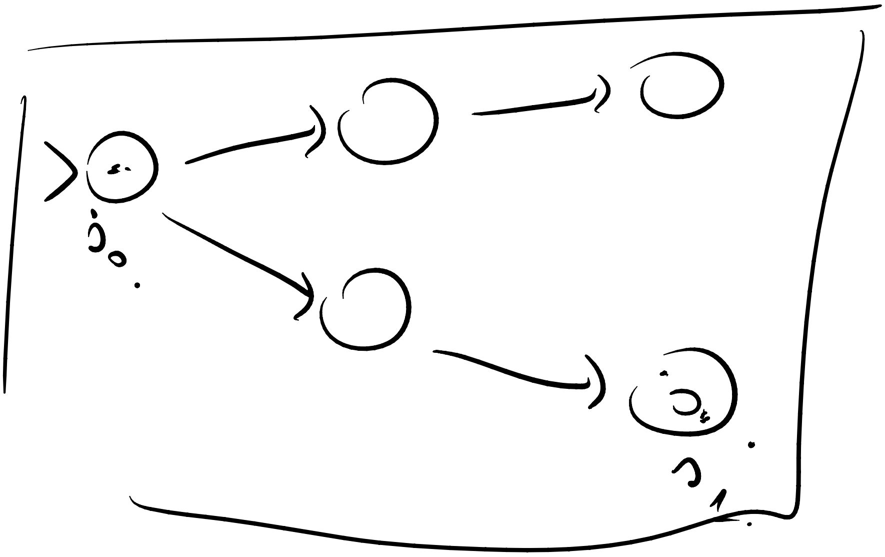
1. $(M ÷ e)$. ✓
2. $(M ÷ 1)$. ✓.

TA: $x_i \in L$ Y.

$x_i \notin L$ ∞.

9. $$L = \{ \rho(M), \underbrace{\underline{\boxed{M \in AFD}}} / \underline{\underline{|L(M)| = \infty}} \}.$$

$\underline{\in TJ.}$



$\underline{\neq \infty}$

$\underline{= \infty}$

10. $L = \{ \rho(M) /$ M este un AFD $\}$

11. $L = \{ \rho(M) /$ M are 3 stări $\}$

12. $L = \{ \rho(M) / \forall \in \Sigma^*,$ M face cel mult 3 tranziții cu

cel mult 3 tranziții cu
w la intrare $\}$

13. $L = \{ \rho(M) / \exists\, w$, M nu se oprește cu w la intrare $\}$

14. Fie L un limbaj format doar din stringul s. s are valoarea 1 dc. extratereștrii există, și 0 dc. nu există. Este L TD?